Flagging and Trust Mechanism

Rules, weights, formulas, examples and the pickup decline rule for the Agrivive surplus vegetable marketplace

Prepared for the Agrivive research documentation. All point values, limits and time windows are initial team settings for the pilot; the cited work supports the criteria behind them, not the numbers.

1. What Flagging and Trust Are, and Why Agrivive Needs Them

Flagging is a tool that lets users, or the system itself, mark a behaviour as a possible problem so that the platform can review it [1]. A **trust mechanism**, also called a reputation system, collects information about past behaviour and turns it into a signal that strangers can use when deciding whether to deal with each other [2], [3]. Buyers, sellers and platform administrators all depend on these tools. Large marketplaces already use them. Shopee, for example, tracks seller cancellations and late shipments and gives penalty points when the rates are too high [5], [6]. Too Good To Go reviews its partner stores by customer rating, cancellation rate and refund rate [7], [8]. These tools solve a basic problem of online trade. Buyers and sellers do not know each other, so one side can act badly without cost, and honest users cannot tell who is reliable [3], [4].

In Agrivive this problem is sharper because the goods are perishable surplus vegetables. A buyer who reserves and does not collect wastes the produce, and a seller who cancels or hands over spoiled produce wastes the buyer's time. Agrivive therefore adds up verified events into one trust score and links the score to clear consequences that grow step by step, which follows the idea of graduated sanctions [9]. A single report is treated as an allegation until independent users agree [1], and every penalty can be explained, disputed and reviewed, which follows the principle of due process for automated decisions [10], [11]. The next sections show how the mechanism works, the events and weights, the formulas, the pickup decline rule, worked examples for a buyer and a seller, and the source of each element.

2. How the Mechanism Works, Step by Step

1. An event happens (for example a reservation expires or an order is cancelled), or a user submits a report.
2. The system checks that the event is **verified** and the **user's own fault**. A single report is saved with 0 points. A second independent report on the same reason confirms it.
3. The system checks the conditions in Tables 1 and 2 (for example a written reason, Away mode, or a confirmed excuse).
4. If the event counts, its weight is added to the trust score S (Equation 1).
5. For a seller, the system also calculates the fault rate F (Equation 2).
6. The system finds the level in Table 3. A consequence starts when a new event moves the user *up* into a higher level. If the score falls later, the level falls without a new penalty.
7. The user receives a plain-word notice that states the event, the rule, how long it lasts, and how to avoid it next time.
8. Within 7 days the user may dispute the event. If it is upheld, the event is removed and the score is recalculated.

3. Verified Events and Weights

An event adds to the trust score only after the system or two independent reporters have verified it, and only if it was the user's own fault. **Never counted:** a single report alone, a cancellation or no-show caused by the other party, an excused event, and an event that was overturned on dispute.

Table 1. Verified events for sellers

Verified event	Weight (w)	Condition	Source
Unanswered buyer inquiry (48 hours)	1	Reminder at 12 h and warning at 24 h carry 0 points. The clock stops while the seller's Away mode is on.	Team design; [5]
Seller cancels a confirmed order	2	None	[5], [7]
Low rating (1 or 2 stars) on a completed order	1	Counts only if the buyer wrote a reason, or it is the 2nd low rating in 30 days.	[2], [7]
Confirmed wrong listing information	2	Two different reporters agree.	[1], [5]
Confirmed inappropriate conduct	2	Two different reporters agree.	[1], [3]
Produce declined at pickup, confirmed as seller fault	2	Valid reason with photo; seller accepts or admin upholds (Section 7).	[1], [7]
Confirmed QR misuse	3 + immediate block	Serious cases skip the steps.	[9]

Table 2. Verified events for buyers

Verified event	Weight (w)	Condition	Source
Reservation expired without pickup	2	Weight is 0 if the buyer's excuse is confirmed.	[7], [9]
Late cancellation (within 3 hours of the pickup slot)	1	None	[8]
Declined at pickup without a valid reason	1	Reason not in Table 8, or admin sides with the seller. Same as a late cancellation.	[3], [8]
False or reused decline photo	2	Admin confirms. Counted as confirmed conduct.	[1], [3]
Confirmed inappropriate conduct	2	Two different reporters agree.	[1], [3]
Confirmed QR misuse	3 + immediate block	Serious cases skip the steps.	[9]

4. Consequence Levels

Both roles use the same score bands, so the rules are easy to understand and equal for everyone. Penalties grow step by step, and serious cases can skip steps [9].

Table 3. Trust levels and consequences

Score (S)	Level	Buyer consequence	Seller consequence
0–1	Good standing	None	None
2–3	Notice	Plain-word notice: what happened, which rule, how to avoid it.	Same
4–5	Warning	Warning is shown at reservation.	Warning is shown in the dashboard.

Score (S)	Level	Buyer consequence	Seller consequence
6–7	Pause 24 hours	New reservations off for 24 hours. Browsing and messages still work.	New listings off for 24 hours; no Priority Boost. Needs F ≥ 10%.
8–11	Pause 3 days	New reservations off for 3 days.	New listings off for 3 days. Needs F ≥ 10%.
12 or more	Block	Account blocked. The user may ask for a review.	Account blocked. The user may ask for a review. Needs F ≥ 10%.

For sellers, a Pause or Block needs both the score and a fault rate (F) of at least 10%. A seller whose rate is lower, or who has fewer than 10 orders, stays at Warning and is marked "Under watch". Confirmed QR misuse still blocks at once.

5. Formulas

5.1 Trust score

The trust score of a user *u* is the sum of the weights of that user's verified events in the last 90 days. Old events leave the score 90 days after they happened, so a user can recover by behaving well. Adding many small signals so that one event does not decide a person follows how reputation systems work [2].

$$S_u = \Sigma\, w_e \text{ (for all verified events } e \text{ in the last 90 days)} \quad (1)$$

5.2 Seller fault rate

Counting only events is unfair to large sellers, because a seller with many orders will naturally have more faults. Marketplaces such as Shopee therefore use rates, and only apply them when there are enough orders [5], [6]. Agrivive uses the rate over 90 days, and only when the seller has at least 10 orders. If F ≥ 10%, a Pause or Block may apply. If F ≥ 15%, the seller gets a "Needs review" flag with 0 points and cannot use Priority Boost or earn the Trusted Seller badge.

$$F = (\text{seller cancellations} + \text{confirmed wrong listings} + \text{upheld declines at pickup}) \div (\text{completed} + \text{cancelled} + \text{declined orders}) \quad (2)$$

5.3 Seller rating

A new seller with two perfect reviews should not rank above a seller with many good reviews. A Bayesian average pulls ratings with few reviews toward the marketplace average [12].

$$R = (C \cdot m + \Sigma\, r) \div (C + n) \quad (3)$$

Here *m* is the marketplace average rating, *C* is the number of "phantom" reviews (set to 5), Σ*r* is the sum of the seller's ratings, and *n* is the number of reviews. For m = 4.0, a new seller with two 5-star reviews has R = (5×4.0 + 10) ÷ (5 + 2) = 4.29. A seller with 40 reviews averaging 4.6 has R = (20 + 184) ÷ (5 + 40) = 4.53. The second seller reaches the Trusted Seller limit of 4.5 and the first does not. A Trusted Seller also needs S = 0 over 90 days, at least 20 completed orders, and F below 15% [3], [4], [7].

6. Fairness Controls and Report Rules

Rules that punish must also protect users from wrong penalties. People need understandable notice of automated decisions and a chance to be heard [10], [11]. Agrivive therefore adds the controls below.

Table 4. Controls that protect users from wrong penalties

Control	Rule
Away mode (seller)	The seller switches it on for sickness, harvest or travel. The 48-hour reply clock stops, and buyers can see the stall is away. Limit: 14 days per switch.
Excuse (buyer)	After an expired reservation the buyer has 24 hours to press "I had a problem" and give a reason (rain, sick, no ride, other). The seller or an admin confirms it. If confirmed, the weight becomes 0. Limit: once per 30 days.
Dispute	Any counted event can be disputed within 7 days with a written reason. An admin reads the order history. If upheld, the event is removed and the score is recalculated.
Review after a penalty	A user under Pause 3 days or Block can ask for a review.
Report rules	A report is an allegation worth 0 points. Two different people must agree. Details must be at least 20 characters. One report per order per person. A report is ignored while the reported person has an open report against the reporter for the same order. The reporter is told the result.
Report-pair check	If the same two people confirm reports against the same person twice in 90 days, that pair no longer counts alone. A third reporter or an admin review is needed.

7. Pickup Inspection and Decline Rule

7.1 Why this rule is needed

Without a rule, two kinds of unfairness appear. A seller who hands over spoiled produce would carry no penalty, because the order was never completed. A buyer could also decline good produce, for example after changing their mind, and the seller would lose the sale with no protection. The rule therefore asks for evidence, separates valid reasons from invalid ones, and gives each side a way to answer. This applies the point that a bare flag is too blunt and can be abused [1], that feedback can be retaliatory [3], and that automated penalties need notice and a chance to be heard [10], [11].

7.2 Valid and invalid reasons

Table 8. Reasons for declining at pickup

Reason	Valid?	Why
Spoiled, rotten or moldy produce	Yes	Not fit to eat. Similar to the "unsafe to eat" case in Too Good To Go's refund measure [7].
Wilted or damaged far beyond what the listing described	Yes	The listing was wrong.
Wrong item	Yes	The listing does not match what was given.
Weight or quantity short by more than 10%	Yes	The listing does not match. The 10% is a team setting.
Signs of contamination or pests	Yes	Food safety.
Size, shape or small blemishes when the listing says "surplus" or "imperfect"	No	The buyer accepted this when reserving.
Changed mind, or no longer wants it	No	Buyer-side reason. Treated as a late cancellation.
Arrived after the pickup slot	No	Treated as an expired reservation.

7.3 Step-by-step process

1. The buyer inspects the produce **before** the QR code is scanned to complete the pickup.

2. If the buyer declines, they press "Decline at pickup", choose a reason from Table 8, and take a photo **inside the app**. The app records the time. The photo is required, because it stops reused or old photos.
3. The seller has 24 hours to **accept** or **dispute**. The seller may add their own photo.
4. If the seller accepts, or does not answer in 24 hours, the decline is treated as accepted.
5. If the seller disputes, an admin reads both photos and the listing, and decides within 3 days. Until then nobody gets points.
6. The order closes as "Declined at pickup". The declined produce is not put back into stock, because it cannot be resold safely. No charge applies to the declined items.
7. Both users get a plain-word notice. Either can still dispute the result within 7 days (Table 4).

7.4 Outcomes, weights and consequences

Table 9. Result of a decline at pickup

Result	Seller points	Buyer points	Condition
Valid reason; seller accepts (or no answer in 24 h)	+2	0	Buyer photo is required.
Valid reason; seller disputes; admin sides with buyer	+2	0	Admin reads both photos and the listing.
Reason not valid under Table 8, or admin sides with seller	0	+1	Counted as a late cancellation.
Buyer's photo shown to be false or reused	0	+2	Counted as confirmed conduct. Needs admin confirmation.
Buyer declines only part of the order	As above, once per order	As above	Only the declined items are removed. One event per order.

A seller fault at pickup gets 2, the same as a seller cancellation, because the buyer made the trip and the produce was wasted. A buyer's invalid decline gets 1, the same as a late cancellation, because the buyer did show up and the seller can still resell the produce. Upheld declines also count in the seller fault rate (Equation 2).

8. Worked Examples

8.1 Buyer example: "Ana"

Ana's score is the sum of her verified events inside the 90-day window. Events that were the seller's fault, excused, or only reported once add 0.

Table 5. Ana's events and score

Day	Event	Counted?	w	Score S	Level and result
1	Reservation expires.	Yes	+2	0 + 2 = 2	Notice
10	Cancels 2 hours before the slot.	Yes (late cancel)	+1	2 + 1 = 3	Notice
20	Reservation expires.	Yes	+2	3 + 2 = 5	Warning
25	Seller closed the stall.	No (seller fault)	0	5	No change
35	Expires because of heavy rain. Ana presses "I had a problem" and the seller confirms.	No (excused)	0	5	No change

Day	Event	Counted?	w	Score S	Level and result
38	Cancels 1 hour before the slot.	Yes	+1	5 + 1 = 6	Pause 24 hours
45	Seller A reports her for conduct.	No (one report)	0	6	Allegation only
46	Seller B reports the same reason.	Yes (confirmed)	+2	6 + 2 = 8	Pause 3 days
60	Reservation expires.	Yes	+2	8 + 2 = 10	Same band, no new penalty
70	Reservation expires.	Yes	+2	10 + 2 = 12	Block

Check at day 70: S = 2 + 1 + 2 + 1 + 2 + 2 + 2 = **12**. The seller-caused event (day 25), the excused event (day 35) and the single report (day 45) add 0. Without the excuse on day 35, Ana would have reached 12 at day 60. At day 70 Ana can ask for a review, and an admin can remove any event that was wrongly counted.

8.2 Seller example: "Mang Tonyo"

Table 6. Mang Tonyo's events and score

Day	Event	Counted?	w	Score S	Level and result
1	Inquiry unanswered for 48 h (reminder at 12 h and warning at 24 h carry 0 points).	Yes	+1	1	Good standing
8	Buyer gives 2 stars and writes a reason.	Yes	+1	2	Notice
15	Cancels a confirmed order (out of stock).	Yes	+2	4	Warning
18	Buyer gives 1 star with no reason.	No (needs a reason)	0	4	No change
20	Buyer A reports wrong listing (said 2 kg, got 1 kg).	No (one report)	0	4	Allegation only
22	Buyer B reports the same reason.	Yes (confirmed)	+2	6	Score is at Pause 24 h. F = (1 + 1) ÷ 10 = 20% ≥ 10%, so the pause applies.
23	Answers an inquiry after 30 h.	No (before 48 h)	0	6	24 h warning only
30	Cancels another confirmed order.	Yes	+2	8	F = (2 + 1) ÷ 12 = 25%, so Pause 3 days applies.
50	Two buyers confirm rude messages.	Yes	+2	10	Same band
65	Inquiry unanswered, but Away mode is on.	No (clock stopped)	0	10	No change
75	Cancels a third confirmed order.	Yes	+2	12	F = (3 + 1) ÷ 14 = 28.6%, so Block applies.

Check at day 75: S = 1 + 1 + 2 + 2 + 2 + 2 + 2 = **12**. The no-reason rating, the single report, the answered inquiry and the Away-mode inquiry add 0.

Why the rate matters. Take a large seller who also reaches S = 12, but from 3 faults in 500 orders. Their rate is 3 ÷ 500 = 0.6%, below 10%. They stay at Warning with an "Under watch" mark and are not paused or blocked. A small seller with many faults per order is stopped, as Mang Tonyo was. The same score therefore leads to different outcomes, which is fairer to sellers of different sizes.

8.3 Decline at pickup examples

Example A: valid decline, seller accepts. Mang Tonyo has S = 8 at day 40. Ana picks up spinach, finds it rotten, takes an in-app photo and declines. Tonyo accepts. Seller: S = 8 + 2 = **10**, the same band (8–11), so no new penalty. His orders go from 12 to 13, and F = (2 cancels + 1 wrong listing + 1 upheld decline) ÷ 13 = 4 ÷ 13 = **30.8%**, so the rate gate stays met. Ana gets 0 points.

Example B: invalid reason. Ana (S = 5) arrives and declines tomatoes because "they look small". The listing said "surplus produce". Size is not a valid reason, so it counts as a late cancellation. Ana: S = 5 + 1 = **6**, which moves her into **Pause 24 hours**. Tonyo gets 0 points and can resell the produce.

Example C: seller disputes. Ana declines for "wilted" with a photo. Tonyo disputes and adds his own photo. The admin sees that Ana's photo shows mold, and Tonyo's photo is of a different crate. The admin sides with Ana: Tonyo's S goes up by 2 and Ana gets 0 points. If the admin had seen that both photos show fresh produce, Ana would get +1 and Tonyo 0.

9. Conclusion

Agrivive's flagging and trust mechanism turns separate events into one score, one set of levels and one set of notices. Reports are treated as allegations until independent users agree. Penalties grow in steps. Sellers are judged by a fault rate and not only by raw counts. The pickup decline rule closes the gap at the moment of handover by asking for evidence from the buyer and giving the seller a way to answer. Excuses, Away mode, disputes and reviews let both sides correct wrong penalties. The worked examples show that the same rules protect honest users and stop repeat violators. All numbers are initial settings for the pilot and should be tuned with real data from the first months of use.

10. Where Each Element Came From

The cited work supports the **criteria** (graduated, corroborated, explained, contestable). It does not give the exact numbers. The numbers are initial team settings.

Table 7. Basis of each element

Element	Basis	Type
Penalties grow step by step; serious cases skip steps	Graduated sanctions [9]; Too Good To Go restricts stores with habitual late cancellations [8]	Research and company terms
Adding weighted events into one score	Reputation systems combine many feedback signals [2]	Research
Counting only the user's own fault	Shopee counts seller-fault events [5], [6]	Company practice
Fault rate with a minimum order count (Equation 2)	Shopee rate-based penalties [5], [6]. The 10% gate, 15% flag and 10-order minimum are team settings	Company practice and team setting
Seller cancellation, rating and issue rates as measures	Too Good To Go partner metrics [7]	Company practice
Report = allegation; two reporters; retaliation guard; report-pair check	Flags are blunt and can be abused [1]; feedback can be retaliatory [3]. The number 2 is a team setting	Research and team setting
Plain notices, dispute, review	Due process for automated decisions [10], [11]	Research
Bayesian rating (Equation 3)	Bayesian average for few reviews [12]. C = 5 is a team setting	Technical source and team setting
Trusted Seller badge	Reputation raises trust and price [3]; institution-based trust [4]; Local Hero tier [7]	Research and company practice

Element	Basis	Type
Away mode, excuse, 14-day, 24-hour and once-per-30-day limits	Team design to prevent unfair penalties, guided by due process [10], [11]. No study gives these numbers	Team design
Required in-app photo; reason from a list (Section 7)	A bare flag is too blunt [1]. Too Good To Go counts customer-reported issues as a store measure [7]	Research and company practice
Seller can accept or dispute; admin decides (Section 7)	Notice and a chance to be heard [10], [11]. Dispute path as in Shopee's returns [5]	Research and company practice
Invalid decline treated as a late cancellation	Late-cancel limit [8]	Company terms
Weights, level thresholds, 90-day window, 3-hour limit, 10% weight tolerance, 24 h seller window, 3-day admin decision	Team settings. No study gives these values	Team setting

Limits to state in the document

Away mode, excuse, dispute, the fault-rate gate, the Trusted Seller badge and the pickup decline flow are design proposals. They are not built into the system yet. References [5]–[8] are company or seller-tool pages, not peer-reviewed research, and Shopee's thresholds differ by country and over time, so cite them as "reported by". References [1]–[4] and [9]–[11] should be checked against the originals for volume and page numbers before submission.

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